

Certificate of Analysis

PRODUCT DETAILS

Name

Product Name

General Exosomes (RejuvEXO)

Batch

Manufacturing Date

Expiry Date

Dosage Form

Clear-colourless solution

Storage Temperature

4 – 8°C

TEST	SPECIFICATION	TEST RESULTS	REMARKS
Infectious Diseases Screening	HBsAg; HCV Ab; HIV1/2 Ag/Ab; CMV Ab; RPR	Not Detected / Non-Reactive	QUALITY CONTROL APPROVED
Sterility	No cloudiness of the test media	Negative	QUALITY CONTROL APPROVED
Bacterial Endotoxin Test (LAL)	Endotoxin sensitivity 0.125 EU/mL	Negative	QUALITY CONTROL APPROVED
Mycoplasma	Mean of absorbance <0.05	Negative	QUALITY CONTROL APPROVED
Simple Western Blot	Positive Marker: Tetraspanin – CD9, CD63	Detected	QUALITY CONTROL APPROVED
	Cytosolic – TSG101 Negative Marker: Cellular – Calnexin	Trace marker detected	
Nanoparticle Tracking Analysis (NTA)	Exosome particle size range 30nm – 200nm	10 x 10 ⁹ particles/mL	QUALITY CONTROL APPROVED

The cell product delivered by Malaysian Genomics Resource Centre Berhad is manufactured and tested for research purpose only. Product is processed in a facility operating under current Good Manufacturing Practice (cGMP) guidelines in accordance with PIC/S standards.

Storage recommendation:

- 1) Refrigerate the vials in 4-8°C at all times upon received
- 2) Keep away the vials from direct heat or sunlight
- 3) Avoid prolong exposure in room condition
- 4) It is recommended to place the vials in freezer for prolonged storage

Authorised Signature:

Quality Control Department

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Supporting Information

STERILITY TEST ^{6, 8}

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.
Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

BACTERIAL ENDOTOXIN TEST (LIMULUS AMOEBOCYTE LYSATE, LAL) TEST ^{3,6}

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

MYCOPLASMA TEST ^{4,6}

Negative findings for mycoplasma species *M. hyorhinis*, *M. arginini*, *M. fermentans*, *M. orale*, *M. Pirum*, *M. hominis*, *M. salivarium* and *A. laidlawii*.

SIMPLE WESTERN BLOT

Marker	Positive Marker		Negative Marker	
	Tetraspanin		Cytosolic	Cellular
	CD9	CD63	TSG101	Calnexin
Result	+	+	+	*

'+' indicates the presence of the marker in the cell.
'*' indicates the trace of the marker detected in the cell

NANOPARTICLE TRACKING ANALYSIS (NTA) ^{7,8}

Established method for the direct and real-time visualization and analysis of the nanoparticles size distribution and concentration in liquids.



REFERENCES

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4. US Pharmacopoeia <63> Mycoplasma Tests
5. CH Harmonised Tripartite Guideline. Specifications: Test Procedures and Acceptance Criteria for Biotechnological/ Biological Products Q3B.
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8. Dae Hyeon Ha, Hyun-Kwon Kim, Joong Lee, Hyuck-Ngon Kwon, Gyeong-Hun Park, Seon Hwang Yang, Jee Yoon Jung, Hwang Choi, Jun Ho Lee, Sumi Sung, Yong Weon Yi, Byung Seung Cho, 2020, Mesenchymal Stem/Stromal Cell-Derived Exosomes for Immunomodulatory Therapeutics and Skin Regeneration.

